

Event Listeners and More Events

1. Event Listeners

JavaScript offers three ways to add an Event Listener to a DOM element.

- Inline event listeners
- onevent listeners
- addEventListener()

1.1 Inline event listeners

```
<button onclick="greeting()">Greet</button>
function greeting() {
  console.log("Hi Rahul");
}
```

1.2 onevent listeners

```
<button id="greetBtn">Greet</button>
let greetBtnEl = document.getElementById("greetBtn");

greetBtnEl.onclick = function() {
  console.log("Hi Rahul");
};
```

1.3 addEventListener()

It is a modern approach to add an event listener.

Syntax:

```
element.addEventListener(event, function);
element - HTML element
```

event - event name
function - Callback function

```
<button id="greetBtn">Greet</button>
let greetBtn = document.getElementById("greetBtn");

greetBtn.addEventListener("click", function() {
  console.log("Hi Rahul");
});
```

2. Operators

2.1 Comparison Operators

Operator	Usage	Description
Equal (==)	$a == b$	returns true if both <i>a</i> and <i>b</i> values are equal.
Not equal (!=)	$a != b$	returns true if both <i>a</i> and <i>b</i> values are not equal.
Strict equal (===)	$a === b$	returns true if both <i>a</i> and <i>b</i> values are equal and of the same type.
Strict not equal (!==)	$a !== b$	returns true if either <i>a</i> and <i>b</i> values are not equal or of the different type.
Greater than (>)	$a > b$	returns true if <i>a</i> value is greater than <i>b</i> value.
Greater than or equal (>=)	$a >= b$	returns true if <i>a</i> value is greater than or equal to <i>b</i> value.
Less than (<)	$a < b$	returns true if <i>a</i> value is less than <i>b</i> value.

Less than or equal (`<=`) `a <= b` returns true if *a* value is less than or equal to *b* value.

2.2 Logical Operators

Operator	Usage	Description
AND (<code>&&</code>)	<code>a && b</code>	returns true if both <i>a</i> and <i>b</i> values are true.
OR (<code> </code>)	<code>a b</code>	returns true if either <i>a</i> or <i>b</i> value is true.
NOT (<code>!</code>)	<code>!a</code>	returns true if <i>a</i> value is not true.

3. More Events

Events are the actions by which the user or browser interact with HTML elements.

There are different types of events.

- **Keyboard Events**
- Mouse Events
- Touch Events, and many more.

3.1 Keyboard Events

Keyboard Event is the user interaction with the keyboard.

The keyboard events are

- `keydown`
- `keyup`

3.1.1 Keydown event

The

`keydown` event occurs when a key on the keyboard is pressed.

Syntax:

```
element.addEventListener("keydown", function);  
let inputEl = document.createElement("input");
```

```
function printKeydown() {  
  console.log("key pressed");  
}
```

```
inputEl.addEventListener("keydown", printKeydown);  
document.body.appendChild(inputEl);
```

3.1.2 Keyup event

The

`keyup` event occurs when a key on the keyboard is released.

Syntax:

```
element.addEventListener("keyup", function);
```

3.2 Event Object

Whenever an event happens, the browser creates an event object.

It consists of information about the event that has happened.

It consists of many properties and methods.

- type
- target
- key
- timeStamp
- stopPropagation, and many more.

3.2.1 Properties & Methods

For any event, event-specific properties and methods will be present.

For Example,

The

`keydown` event has `key` property, whereas the `onclick` event doesn't have it.

`event.type`

The

`event.type` property contains the type of event occurred like `click`, `keydown`, etc.

```
let inputEl = document.createElement("input");
```

```
function printKeydown(event) {  
  console.log(event.type); // keydown  
}
```

```
inputEl.addEventListener("keydown", printKeydown);  
document.body.appendChild(inputEl);
```

`event.target`

The `event.target` property contains the HTML element that triggered the event.

```
let inputElement = document.createElement("input");
```

```
function printKeydown(event) {  
  console.log(event.target); // <input></input>  
}
```

```
inputElement.addEventListener("keydown", printKeydown);
```

```
document.body.appendChild(inputElement);
```

event.key

The **event.key** property contains the value of the key pressed by the user.

```
let inputElement = document.createElement("input");
```

```
function printKeydown(event) {  
  console.log(event.key);  
}
```

```
inputElement.addEventListener("keydown", printKeydown);  
document.body.appendChild(inputElement);
```

Keyboard key	event.key value
Enter	Enter
a	a
A	A
1	1
*	*
<	<